Synoptic Question 5: Geostationary Communications Satellite

Reading Passage

Read the following article and then answer the questions that follow.

Modern global communications rely heavily on a network of artificial satellites orbiting the Earth. One of the most important types is the geostationary satellite. A satellite in a geostationary orbit remains at a fixed point above the Earth's equator. This is achieved by matching the satellite's orbital period exactly to the Earth's rotational period. This fixed position allows a ground-based antenna to be pointed permanently at the satellite, making it ideal for continuous communication services like satellite television and long-distance telephone calls.

To achieve this orbit, the satellite must be placed at a very specific altitude where the gravitational force provides the exact centripetal force required for it to travel at the necessary speed. The signals transmitted to and from the satellite are high-frequency microwaves, which travel in straight lines and can pass through the Earth's ionosphere with minimal distortion.

A typical communications satellite is powered by large solar panel arrays which convert sunlight into electrical energy. This energy is used to power the onboard electronics and to transmit signals back to Earth. The power of the transmitted signal decreases with distance according to the inverse square law. As the signal passes through the atmosphere, it also experiences some loss of strength, known as attenuation.

Data:

Table 1: Data for Earth and a typical Satellite

Parameter	**Value**
Mass of Earth, M_E	5.97×10^{24} kg
Radius of Earth, R_E	6.37×10^{6} m
Gravitational constant, G	6.67×10^{-11} N m^2 kg^{-2}
Mass of satellite, m	4500 kg
Frequency of uplink signal	14.0 GHz
Power of transmitted signal from satellite, P_{trans}	2.50 kW
Area of satellite's solar panels	50 m^2
Efficiency of solar panels	28%
Solar radiation intensity at orbit, I_{solar}	1360 W m^{-2}

Questions

(a) For a satellite to be in a geostationary orbit:

(i) State the period of the orbit in seconds.

(ii) State the direction of the orbit relative to the Earth's rotation.

(b) By equating the gravitational force on the satellite with the required centripetal force, show that the radius r of the satellite's orbit is given by the expression:

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

where T is the period of the orbit.

(c) Calculate:

(i) The radius of the geostationary orbit.

(ii) The altitude of the satellite above the Earth's surface.

(iii) The linear speed of the satellite.

(d) The signals used for communication are microwaves.

(i) A 14.0 GHz signal is sent from a ground station to the satellite (the uplink). Calculate the wavelength of this signal.

(ii) Calculate the minimum time delay for a signal to travel from a ground station on the equator to the satellite and back to the ground station.

(e) The satellite transmits a signal with a power of 2.50 kW. This signal spreads out uniformly in a sphere.

(i) Assuming the signal spreads isotropically (uniformly in all directions), calculate the intensity of the signal when it reaches the Earth's surface.

(ii) The signal is received by a parabolic dish on Earth with a diameter of 1.2 m. Calculate the power received by this dish.

(f) The satellite is powered by solar panels.

(i) Calculate the total electrical power generated by the solar panels.

(ii) Suggest why the power generated might be greater than the power required to transmit the signal.

(g) Calculate the gravitational potential energy of the satellite in its orbit. (Gravitational potential at a point is defined as zero at infinity).

Solutions

(a) (i) The period must be the same as the Earth's rotational period, which is 24 hours. $T = 24\text{ h} \times 60\text{ min/h} \times 60\text{ s/min} = 86400\text{ s}$.

(ii) The direction of the orbit must be the same as the Earth's rotation (from west to east).

(b) Gravitational Force = Centripetal Force

$$\frac{GM_E m}{r^2} = mr\omega^2$$

Since angular velocity $\omega = \frac{2\pi}{T}$:

$$\frac{GM_E}{r^2} = r\left(\frac{2\pi}{T}\right)^2 = \frac{4\pi^2 r}{T^2}$$

Rearranging for r^3:

$$r^3 = \frac{GM_E T^2}{4\pi^2}$$

$$r = \sqrt[3]{\frac{GM_E T^2}{4\pi^2}}$$

(c) (i) $r = \sqrt[3]{\frac{(6.67\times10^{-11})(5.97\times10^{24})(86400)^2}{4\pi^2}} = \sqrt[3]{7.53\times10^{22}}$
$r = 4.22\times10^7$ m.

(ii) Altitude $h = r - R_E = (4.22\times10^7) - (6.37\times10^6) = 3.58\times10^7$ m (or 35,800 km).

(iii) Linear speed $v = r\omega = r\frac{2\pi}{T} = (4.22\times10^7)\frac{2\pi}{86400} = 3070\text{ m s}^{-1}$.

(d) (i) $f = 14.0\text{ GHz} = 14.0\times10^9$ Hz. Speed of light $c = 3.00\times10^8\text{ m s}^{-1}$.
Wavelength $\lambda = \frac{c}{f} = \frac{3.00\times10^8}{14.0\times10^9} = 0.0214$ m.

(ii) The signal travels from ground to satellite and back, covering a total distance of $2h$.
Total distance $= 2\times(3.58\times10^7) = 7.16\times10^7$ m.
Time delay $t = \frac{\text{distance}}{\text{speed}} = \frac{7.16\times10^7}{3.00\times10^8} = 0.239$ s.

(e) (i) The signal power spreads out over the surface area of a sphere with radius equal to the satellite's altitude.
Surface area $A = 4\pi h^2 = 4\pi(3.58\times10^7)^2 = 1.61\times10^{16}\text{ m}^2$.
Intensity $I = \frac{P_{trans}}{A} = \frac{2500}{1.61\times10^{16}} = 1.55\times10^{-13}\text{ W m}^{-2}$.

(ii) Radius of dish $r_{dish} = 0.6$ m. Area of dish $A_{dish} = \pi r_{dish}^2 = \pi(0.6)^2 = 1.13\text{ m}^2$.
Power received $= I\times A_{dish} = (1.55\times10^{-13})\times1.13 = 1.75\times10^{-13}$ W.

(f) (i) Power incident on panels $= I_{solar}\times\text{Area} = 1360\times50 = 68000$ W.
Electrical power generated = Power incident × Efficiency $= 68000\times0.28 = 19040$ W $=$ 19.0 kW.

(ii) The extra power is needed to run all the satellite's onboard electronic systems (computers, sensors, cooling systems) and to charge batteries for times when the satellite might be in the Earth's shadow (eclipse).

(g) Gravitational potential energy $U = -\frac{GM_E m}{r}$.

$$U = -\frac{(6.67\times10^{-11})(5.97\times10^{24})(4500)}{4.22\times10^7}$$

$$U = -4.24\times10^{10}\text{ J}$$

[Diagram of a satellite in geostationary orbit above the Earth's equator]

Figure 1: Geostationary Satellite Orbit

Cambridge International AS & A Level Physics (9702)

Paper 4: A Level Structured Questions

Variant 2 – Higher Difficulty

Topic Focus: Satellite Motion and Dynamics Only

Instructions

- Answer all questions.
- Show all relevant working.
- Marks are shown in square brackets.

Question Paper (Variant 2)

Question 1: Circular Orbit Dynamics

Comprehension passage: A mission brief states that a 900 kg Earth-observation satellite travels in a circular path at a distance of 7.20×10^6 m from Earth's center. The same brief lists $G = 6.67 \times 10^{-11}$ N m^{2}kg^{-2}, Earth mass 6.00×10^{24} kg, and Earth radius 6.40×10^6 m. Use these values as needed.

(a) Show that gravitational force provides the centripetal force and derive $v = \sqrt{GM/r}$. [4]

(b) Extract the required numerical values and calculate the orbital speed and orbital period. [4]

(c) Determine the altitude of the satellite and compare the local gravitational field strength with g at Earth surface. [4]

(d) Open-ended: explain why a low-Earth-orbit satellite does not require engine thrust continuously to remain in orbit, but does require occasional propulsion corrections. [4]

Question 2: Orbit Transfer and Energy Budget

Comprehension passage: A transfer plan reports that a 1200 kg satellite is raised from a circular orbit of radius 8.00×10^6 m to a new circular orbit of radius 1.60×10^7 m. The mission data sheet uses the same constants $G = 6.67 \times 10^{-11}$ N m^{2}kg^{-2} and Earth mass 6.00×10^{24} kg.

(a) Derive an expression for total mechanical energy E in a circular orbit and show that $E = -GMm/(2r)$. [4]

(b) Extract the orbital radii from the passage and calculate the change in total mechanical energy of the satellite–Earth system. [4]

(c) Determine the change in gravitational potential energy and the change in kinetic energy between the two circular orbits. [4]

(d) Open-ended: discuss why an efficient transfer generally uses two engine burns (e.g. transfer orbit) rather than one continuous burn all the way. [4]

Question 3: Geostationary Orbit and Dynamics Constraints

Comprehension passage: A communications company requires a satellite that appears fixed over one longitude. Engineering notes state Earth rotates once every 8.64×10^4 s. Use this period together with standard Earth constants from earlier questions.

(a) State the conditions for a truly geostationary orbit. [3]

(b) Show that geostationary orbital radius is given by

$$r = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3}$$

and calculate its value. [5]

(c) Hence calculate the orbital altitude above Earth's surface. [2]

(d) Open-ended: explain, using dynamics and mission requirements, why a satellite with the correct period but inclined orbit is not geostationary from a ground observer's perspective. [4]

Question 4: Re-entry, Drag and Stability (Open-ended Dynamics)

Comprehension passage: During re-entry testing, a capsule of mass 2400 kg descends vertically and sensors record an average upward drag of 3.0×10^4 N close to parachute deployment. Assume gravitational field strength $g = 9.81\,\mathrm{ms}^{-2}$.

(a) Explain, in terms of momentum change and force, why deceleration increases rapidly as atmospheric density increases. [4]

(b) Describe the role of drag force in converting mechanical energy during re-entry. [3]

(c) Extract the numerical data from the passage and calculate the capsule acceleration. [3]

(d) Open-ended: discuss two design features that improve re-entry stability and safety, linking each feature to the underlying dynamics. [4]

Sample Solutions (Variant 2)

Q1 Sample Solution

(a) For circular motion, centripetal force needed is mv^2/r. Gravitational force is GMm/r^2. Equating: $GMm/r^2 = mv^2/r$, so $v = \sqrt{GM/r}$.

(b)

$$v = \sqrt{\frac{(6.67 \times 10^{-11})(6.00 \times 10^{24})}{7.20 \times 10^6}} = 7.46 \times 10^3 \text{ m s}^{-1}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi(7.20 \times 10^6)}{7.46 \times 10^3} = 6.06 \times 10^3 \text{ s} \, (\approx 101 \text{ min})$$

(c) Altitude $h = r - R_E = 8.0 \times 10^5$ m.

$$g_h = \frac{GM}{r^2} = 7.72 \text{ m s}^{-2}$$

So local g is about $7.72/9.81 \approx 0.79$ of surface value.

(d) In orbit, gravity continuously provides required centripetal acceleration, so no continuous tangential thrust is needed. Corrections are required for perturbations (atmospheric drag, Earth's non-uniform field, third-body effects, solar radiation pressure).

Q2 Sample Solution

(a) For circular orbit, $v^2 = GM/r$.

$$E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}\frac{GMm}{r} - \frac{GMm}{r} = -\frac{GMm}{2r}$$

(b)

$$\Delta E = \left(-\frac{GMm}{2r_2}\right) - \left(-\frac{GMm}{2r_1}\right) = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\Delta E = \frac{(6.67 \times 10^{-11})(6.00 \times 10^{24})(1200)}{2}\left(\frac{1}{8.00 \times 10^6} - \frac{1}{1.60 \times 10^7}\right) = 1.50 \times 10^{10} \text{ J}$$

(c)

$$\Delta U = -GMm\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = +3.00 \times 10^{10} \text{ J}$$

$$\Delta K = \Delta E - \Delta U = -1.50 \times 10^{10} \text{ J}$$

(kinetic energy decreases in higher circular orbit).

(d) Two-burn transfer is more fuel-efficient (lower total Δv), allows accurate orbit shaping and phasing, and reduces prolonged thrusting errors.

Q3 Sample Solution

(a) Conditions: circular orbit, period equals Earth's rotation, orbit in equatorial plane, same direction as Earth rotation.

(b) With $mv^2/r = GMm/r^2$ and $v = 2\pi r/T$:

$$\frac{m(2\pi r/T)^2}{r} = \frac{GMm}{r^2} \Rightarrow r^3 = \frac{GMT^2}{4\pi^2} \Rightarrow r = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3}$$

$$r = 4.23 \times 10^7 \text{ m}$$

(c)

$$h = r - R_E = 4.23 \times 10^7 - 6.40 \times 10^6 = 3.59 \times 10^7 \text{ m}$$

(d) If orbit is inclined, satellite appears to move north-south in the sky (analemma/figure-of-eight path) despite correct period. So it is geosynchronous but not geostationary, failing fixed-point ground coverage.

Q4 Sample Solution

(a) As density rises, more air mass is intercepted per second, increasing rate of momentum change and hence drag force $F = dp/dt$. Larger upward resultant force gives higher deceleration.

(b) Drag does negative work on capsule, reducing kinetic (and some gravitational potential) energy. Mechanical energy is transferred mainly to thermal/internal energy of gas and heat shield.

(c) Weight $mg = (2400)(9.81) = 2.35 \times 10^4\,\mathrm{N}$ downward. Net upward force $= 3.0 \times 10^4 - 2.35 \times 10^4 = 6.46 \times 10^3\,\mathrm{N}$.

$$a = \frac{F_{\text{net}}}{m} = \frac{6.46 \times 10^3}{2400} = 2.69\,\mathrm{m\ s^{-2}}\ \text{upward}$$

(d) Example features (any two, developed):

- blunt-body shape to increase drag and reduce peak heating/deceleration,
- center-of-mass offset for passive aerodynamic stability,
- controlled attitude thrusters to manage angle of attack,
- staged parachute deployment to limit jerk and dynamic loads.

Mark Scheme (CIE 9702 P4 Style – Variant 2)

Question 1 **[16]**

- (a) Uses gravitational = centripetal relation and derives $v = \sqrt{GM/r}$. [4]
- (b) Correct v and T with valid method and units. [4]
- (c) Correct altitude and field-strength comparison with Earth surface value. [4]
- (d) Relevant dynamics explanation: no continuous thrust + justified perturbation corrections. [4]

Question 2 **[16]**

- (a) Correct derivation of $E = -GMm/(2r)$. [4]
- (b) Correct ΔE with sign and unit. [4]
- (c) Correct ΔU and ΔK and interpretation of reduced kinetic energy at higher orbit. [4]
- (d) Two-burn transfer discussion linked to Δv efficiency and mission control accuracy. [4]

Question 3 **[14]**

- (a) States all required geostationary conditions. [3]
- (b) Correct derivation and numerical orbital radius. [5]
- (c) Correct altitude from Earth surface. [2]
- (d) Clear explanation of inclined-orbit apparent motion and consequence for fixed ground coverage. [4]

Question 4 **[14]**

- (a) Momentum-change explanation linked to force and density increase. [4]
- (b) Correct energy-transfer pathway during drag deceleration. [3]
- (c) Correct resultant-force method and acceleration value with direction. [3]
- (d) Two valid safety/stability features with dynamics linkage. [4]

Total: 60 marks